

No.	Topics	Page No.
1	GCP Certifications	3
2	Basic of Cloud Computing	3
3	What is Iaas, Pass and Saas cloud service	3
4	Concepts of Zones, Regions and Multi-Region	4
5	GCP Resource Hierarchy	5
6	Google Cloud Compute Engine	7
7	Google Cloud Compute Engine pricing & disk types	10
8	Changing the machine type of a VM instance	13
9	Resize your persistent disk of VM	14
10	What is Public image, Custom Image & Machine Image	15
11	What is Cloud Storage in GCP	17
12	Google Cloud Storage Types and their Characteristics	18
13	Google Cloud Storage pricing and the concept of Bucket & Objects	20
14	Complete demo on Cloud Storage using GUI and Cloudshell	21
15	Object Versioning and Lifecycle Management	23
16	How to Create Signed URL	26
17	Google Cloud VPC Network	27
18	VPC Peering	28
19	Cloud NAT Gateway	29
20	Load Balancer in GCP	30
21	Identity and Access Management	32
22	Identity Aware Proxy- Demo	37
22	App Engine- Standard & Flexible	38

23	Cloud Run	40
24	Google Cloud Function	41
25	Google Cloud SQL	43
26	Cloud Spanner	45
27	Cloud Bigtable	46
28	Cloud Memorystore	47

Lecture 1. GCP Certifications

Certification Path

- Foundational certification - Cloud Digital Leader, Duration 90 min, fee \$99
- Associate certification - Associate Cloud Engineer, Duration 2 hr , fee \$125
- Professional certification - Cloud Architect, Cloud developer, Data Engineer, Cloud Network Engineer, Duration 2 hr, fee \$125
- If you fail 1st exam then you have to retake a minimum of 14 days.
If you fail 2nd time then you give a retake after 60 days
- If you fail 3rd time then you gives a retake after 1 year

02. Basics of cloud computing

- Cloud computing is the delivery of computing services:- servers, storage, databases, networking tools & software over the internet.
- Characteristics of cloud computing
 1. On-demand self-service
 2. Broad network access
 3. Resource pooling
 4. Rapid elasticity
 5. Measured services

03. What is Iaas, Pass and Saas cloud service

1. IAAS
2. PAAS
3. SAAS

Types of Clouds / Cloud Deployment Models

1. Public cloud
2. Private cloud
3. Hybrid cloud

Lecture 04- Concepts of Zones, Regions and Multi-Region

- **Regions** are independent geographic areas that consist of zones. they affect pricing, reliability, networking and performance
zonal resource- VM
- **Zone**- A zone is a deployment area for Google Cloud resources within a region. zones should be considered a single failure domain within a region to deploy fault-tolerant applications with high availability and help protect against unexpected failures, deploy your application across multiple zones in a region
Regional Resource- app engine
- **Multi-region** services are designed to be able to function following the loss of a single region.
multi-regional resources- bigquery, bigtable, cloud storage, spanner, datastore, firestore, artifact registry.

Note:- if a single region fails, only customers solely in that region are impacted, customers who have multi-region products are not impacted.

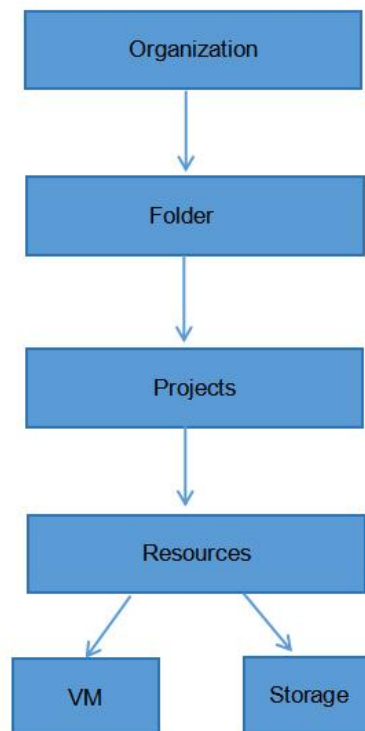
The fully qualified name for a zone is made up of <region><zone>
for eg. zone-a in region us-central1 is us-central1-a

When selecting zones yourself, here are some things to keep in mind

1. Communication within and across regions will incur different costs.
2. Design important Systems with redundancy across multiple zones or regions At Some point in time, your instance might experience an Unexpected failure To mitigate the effect of these possible events, you should duplicate important & systems in multiple Zones and Regions.

Lecture 5: GCP Resource Hierarchy

- The purpose of the Google Cloud Resource Hierarchy is two fold -
 1. Provide a hierarchy of Ownership, Which binds the lifecycle of a resource to its immediate parent in the Hierarchy.
 2. Provide attach points and inheritance for access control and organisation policies.
- GCP Resource hierarchy resembles the file system found in traditional O.S. as a way of organising and managing entities hierarchically. Each resource has exactly one parent.



Projects

- All GCP resources you use are associated with one specific project
- You can track resources and quota
- Enable billing and set budget
- Manage permissions and credentials
- Project is a global entity
- Enable services and API
- Equivalent to an account in AWS and a subscription in Microsoft Azure

Benefits of the Organisation Resource

- With an Organization Resource, projects belong to your organization instead of the employee who created the project. This means that the projects are no longer deleted When an employee leaves the company.
- You Can grant roles at the Organization level, which are inherited by all projects and folders under the Organisation resource for eg. you can grant the network admin role to your networking team at the Organization level, allowing them to manage all the networks in all projects in your company instead of granting them for all individual projects.

Folders

- Folder resources provide an additional grouping mechanism and isolation boundaries between projects.
- Folders can contain other Folders.
- It is optional

Project ID	Globally Unique	Chosen by you	Immutable (when creating - mutable)
Project name	Uniqueness, not req.	Chosen by you	mutable
Project Number	Globally Unique	Assigned by GCP	Immutable

Lecture 07- Google Cloud Compute Engine

- Compute engine is a customizable compute service that lets you create and run virtual machines on Google infrastructure.
- There are two types of machines in GCP
 1. Custom machine: you can take VCPU and memory as per your need and create your own custom machine.
 2. Predefined machines: predefined machine types are pre-built and ready to go configurations of VMS with a specific amount of CPU and memory to start running app quickly.
- Google Compute Engine comes under IAAS.
- A virtual machine is like a physical hardware server but running in the cloud.
- GCP isolates resources and sells them on a Server Basis with different combinations. Google provides VM of all sizes micro to 160 vCPU and 3.75 TB memory.
- You can attach local SSD up to 3 TB and persistent disk up to size of 64 TB, which too encrypted as well
- Google Cloud uses open-source KVM Hypervisor at Backed (AWS=Nitro, Azure=hyper-v)
- Fast Booting time among all cloud providers.
- All VCPU, GCP and GB of memory are charged a minimum of 1 minute after 1 minute, instances are charged in 1 sec increments.
- Types of pre-defined VM/ machine family
 1. General Purpose
 2. Compute Optimize
 3. Memory Optimized
 4. Accelerator
 5. Shared core
- A machine type is a set of virtualised hardware resources available to a virtual machine including the system memory size, vCPU and persistent disk limits.
- In compute engine, machine types are grouped and put inside families on their behaviour & workload.

Compute engine has 3 terms

1. Machine Family- Set of processors and Hardware Configuration
2. Series- Machine families are further classified by Series and Generation

3. Machine type- Every Machine Series has pre-defined machine types that provide a set of resources
4. Customers Using GCE -> PayPal, Twitter, P&G, Airbus, Etsy, SKY

Cost-optimized	Balanced	Scale-out optimized	Memory Optimized	Compute Optimized	Accelerator Optimized
E2	N1,N2,N2D	Tau,T2D	M1,M2	C2, C2D	A2
Day to Day Computing at a Lower Cost	Balanced price/performance across a wide range of VM Shapes.	Best Performance & cost for scale-out Workloads	Ultra-high memory Workloads Offers Highest Memory Config.across VM Families	Ultra-performance compute-intensive workload	Optimized for high-performance computing workload Best for parallel computing workload
- Web Serving - App Serving - Back office pps - Small & Medium Databases - Microservices - Virtual Desktops	- Web Serving - App Serving - Back office apps - Medium-large Databases - Cache - Media/streaming	-Scale-Out Workloads - Web Serving - Containerized Microservices. - Media Transcoding - Large Scale Java Application	- Medium-large in-memory databases such as SAP HANA - In-memory Databases and in-memory analytics -Microsoft SQL Server and Similar Databases	- Compute Bound Workloads. - High-performance Web Serving. Gaming - Ad Serving -High-performance Computing Media	- CUDA enabled ML Training and In Inference - HPC Massive Parallelized Computation

Shared-Core Machines

- This type of VMs intends to timeshare the physical core shared core machine types are a cost-effective option that works well with small or batch workloads that only need to run for a short time.
- It uses partial vCPU that runs on one - hyper thread of the host CPU running your instance.

Sole-Tenant Nodes

- Sole tenant nodes are the servers for the compute engine that are dedicated for the users on priority. The purpose of these nodes is to serve the deployment for BYOL (Bring your own license) applications.
- The sole tenant nodes provide you, access to similar VM Configuration and Machine types just like regular compute instances.

Custom Machine type

- If none of the predefined machine types matches your needs, you can independently specify the number of CPU's and the amount of memory for your instance.
- The memory per vCPU of custom machine types must be between 0.9G18, 3.75&6.5G18 per vCPU.
- The total memory for a custom machine type must be a multiple of 256 MB.
- Above 1, the instance vCPU count must be even, such as 2, 4, 6, 8, 10 and so on.

Lecture 8. Google Cloud compute engine pricing & disk types

Compute Engine Pricing

1. Sustained Use Discount

- Sustained Use discounts are applied automatically to usage within a project Separately for each region, so there is no action required on your part to enable these discounts.
- If you run a VM for more than 25% of a month, you are eligible for a discount between 20-30%.
- Discount is given on vCPU and Memory Usage.
- For general-purpose N2 and N2D predefined and custom machine types and for compute optimized machine types, you can receive a discount of upto 20%.
- For general-purpose N1 predefined and custom machine types Sole-tenant nodes and GPU, you can get a discount of upto 30%.

Limitations

- Sustained use discounts automatically apply to VMs created by GCE or GKE.
- Sustained Use discounts do not apply to VMs created using the app engine environment and dataflow.
- Sustained use discounts do not apply to E2, A2 or Tau T2D machine types.

2. Committed Use Discount

- When you purchase a committed use contract you purchase compute engine resources - Such as vCPUs, memory, GPUs, Local SSDs and Sole-tenant node - at a discounted price in return for committing to paying for those resources for 1 year to 3 years.
- Discount is upto 57% to 70%.
- After purchase, you are billed monthly, whether you are using it or not.
- No upfront cost is required.

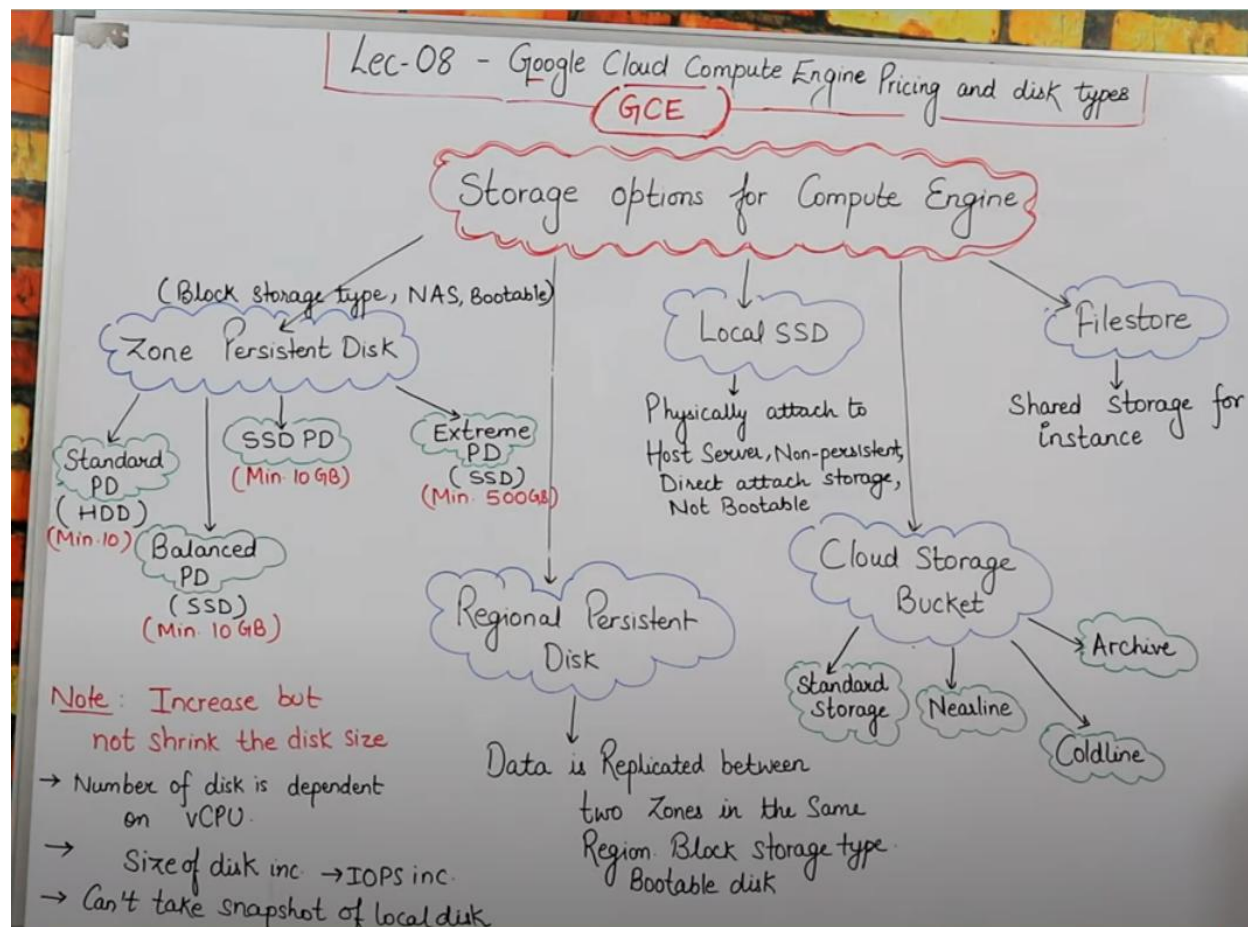
How it works

- For example, if you purchase a Commitment for 18 cores and you run 24 Cores during the month, you will receive a committed Use the discount for 8 Cores The Other 16 Cores will be billed at Standard, non-committed use rates.
- Note-Keep in mind that after you purchase a set amount of Commitments, you are billed for those commitments monthly, even if you don't use them.
- You can't combine committed use discounts and sustained use discounts for the same resources.

3. Preemptible Instances

- Preemptible Instances are available at a much lower price - 60-91% discount, compared to the price of standard VMs.
- However, Google Cloud might stop these instances if it needs to reclaim the compute capacity for allocation to other VM.
- Compute engine always Stops preemptible instances after they run for 24 hours.
- The GCP free tier Credits for the compute engine do not apply to preemptible instances.
- Preemptible instances cannot live to migrate to a regular VM instance or vice versa.
- Not Covered under SLA.
- GCP gives only 30 sec time before reclaiming it.
- Best Suited for Batch processing jobs of Short duration.

Disk Types



- Disk type

1. Zone persistent disk (root storage/ bootable) (block storage)
2. Regional persistent disk (replica create)
3. Local ssd (non-bootable)
4. Cloud storage bucket (non-bootable) (object storage) (not install OS)
5. Filestore (non-bootable)

Lec 13. Changing the machine type of a VM instance

- If your existing machine is not a good fit for the workloads you run on your VM, change the machine type of that VM.
- If your VM does not have a local SSH and is not part of a managed Instance Group (MIG), you can change the machine type of your VM after stopping it.
- If your VM has a local SSD, you cannot stop the VM to change its machine type unless you force it to stop, which causes you to lose all of the data on the local SSD.
- Changing a machine type might also affect the sustained use discounts for that VM.
- You can only change the machine type of a stopped VM you cannot change the machine type of a running VM.
- Take a snapshot before updating the size
- For VMs that do not have a local SSD and are not part of MIG, you can change the machine type without affecting the following resources-
 1. the VMs ssh keys
 2. the VM configuration such as metadata
 3. persistent disk data including installed application and app data.

Lec 14. Resize your persistent disk of VM

- You can increase the size of your persistent disk when your VM instances require additional storage space or higher performance limits.
- You can increase the size of persistent disk you cannot reduce.
- VM using public images automatically resize the boot disk in all other cases. you need to manually resize the boot disk or non-boot disks.
- No need to stop the VM, to resize.

gcloud compute disks resize instance-1 --zone us-central1-a --size 30

Now to resize in Linux VM

```
take aah
sudo sudo
df -th
Lsblk
parted /dev/sda
resize part
1 (partition number)
yes
974? End --> 100%
quit
Now check again
lsblk
```

```
root@instance-1:/home/sakemadhav97# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda          8:0    0   10G  0 disk
├─sda1       8:1    0   9.9G  0 part /
├─sda14      8:14   0    3M   0 part
└─sda15      8:15   0   124M  0 part /boot/efi
```

Lec 15. What is Public image, Custom Image & Machine Image

- You can O.S. system images to create a boot disk for your instance you can use one of the following image types.

Public Image:-

- Public image is provided and maintained by Google, open-source communities and third-party vendors. By default, all Google Cloud projects have access to these images and use them to create instances.

Custom Images- (AMI)

- Custom Images are available only to your cloud project you create a custom image from a boot disk and other images. then use the custom image to create an instance.
- Custom images are free of cost but do incur an image storage charge.
- You can use custom images for the following purpose.
- Import a virtual disk to the compute engine from your on-premises environment or from VMs that are running on your local workstation or on another cloud platform.
- Create an image from the boot disk of your existing VM then use that image to create new boot disks for your instances.
- Copy one image to another image by using the gcloud either the cloud cLI or the API. you can also create an image from a custom image in a different project.

Machine Image: (Multiple Disks)

- Machine image is a compute engine resource that stores all the configuration, metadata, permissions and data from multiple disks of a VM instance you can use a machine image in many system maintenance, backup recovery and instance closing scenarios.
- Machine images are the most ideal resource for the following cases:-
 1. Multiple disk backups
 2. Instance cloning
- Machine Images are suitable for creating backups of all disks that are attached to a VM instances.
- Machine images can be used instance you can create multiple copies of an instance.
- Information stored in a machine image:-
- description, machine type, labels network tages, maintenance policy & instance metadata.
- Machine image stored in a cloud storage bucket.
- Regardless of where the machine image is stored, you can still access the machine image globally.
- Machine images are unchangeable but you can override a few properties.

SCENARIOS	MACHINE IMAGE	PERSISTENT DISK SNAPSHOT	CUSTOM IMAGE	INSTANCE TEMPLATE
Single Disk Backup	YES	YES	YES	NO
Multiple Disk Backup	YES	NO	NO	NO
Differential Backup	YES	YES	NO	NO
Instance Cloning	YES	NO	YES	YES
Base image for Replication	NO	NO	YES	NO

Lec 18. What is Cloud Storage in GCP

- Storage types in Google Cloud
 1. Cloud storage
 2. Persistent Disk
 3. Filestore
- Cloud storage is a service for storing your objects in Google Cloud an object is an immutable piece of data consisting of a file of any size and format.
- You store objects in containers called buckets
- ways to access/interact with cloud storage
 1. Console
 2. gsutil
 3. Client libraries
 4. Rest APIs
- Cloud storage classes
 1. Standard storage
 2. Nearline
 3. Cold line
 4. Archive storage

Lec 19. Google Cloud Storage Types and their Characteristics

- Storage class is a piece of metadata that is used by every object.
- The storage class set for an object affects the object availability and pricing model.
- You can change the storage class of an existing object either by rewriting the object or by using object lifecycle management.
- When you create a bucket, you can specify the default storage class for the bucket if you don't specify, then default storage is standard.

All storage classes provide the following:

- Unlimited storage with no minimum object size.
- worldwide accessibility and worldwide storage locations.
- Low latency.
- High durability (99.999999999).
- Geo redundancy, if the data is stored in a multi-region or dual-region.
- A Uniform experience with cloud storage features, security, tools and APLs.
- Object written to cloud storage must be redundantly stored in at least two different availability zones before the write is acknowledged as successful.
- Max size of an object stored in a cloud bucket is 5 TiB.
- There is a 100 ACL limit per object.
- A multiple upload can have up to 10,000 parts. an individual part has a max size limit of 5 gib objects assembled from a multipart upload are subject to the overall 5TiB size limit.

Standard Storage

- Standard storage is best for data that is frequently accessed (Hot Data) or stored for only brief periods of time.
- When used in a region, standard storage is appropriate for storing data in the same location as GKE cluster or compute engine instances that use the data.
- Regional storage is redundant within a region.
- when used in dual-region improved availability.
- when used in multi-region, ideal for storing data that is accessed around the world such as serving website content, streaming videos executing interactive workloads or serving gaming data or apps.
- multi-region is Geo-redundant stored across 2 or more regions which are 500 miles apart.
- availability of standard storage data is

location type	availability SLA
multi-region	99.95%
dual region	99.95%
Regional	99.90%

Nearline Storage

- Nearline storage is a low-cost, highly durable storage service for storing infrequently accessed data.
- Nearline storage is a better choice than standard storage in scenarios where slightly lower availability, 30-day minimum storage duration and cost for data access are acceptable trade-offs for lowered at-rest storage.
- Nearline storage is ideal for a data plan to read or modify on average once per month.
- Availability of nearline storage data is

location type	availability SLA
multi-region	99.9%
dual region	99.9%
Regional	99.0%

Coldline storage

- Coldline storage provides very low-cost highly durable storage service for storing infrequently accessed data (cold data).
- Coldline storage is ideal for data you plan to read or modify at most once a quarter.
- Availability is the same as Nearline.

Archive storage

- is the cheapest, highly durable storage service for data archiving online backup and disaster recovery.
- unlike the coldest storage service offered by other cloud providers, your data is available within milliseconds, not hours or days.
- archive storage has a higher cost for data access, as well as 367 days min storage duration
- archive storage is the best choice for data that you plan to access less than once a year.
- Availability is same as nearline or coldline.

To maximize the availability of your data

- consider storing your data in a multi-region or dual-region bucket location high availability is a top requirement
- all data is stored geo-redundantly in these locations, which means your data is stored in at least two.
- Geographically separated regions in the unlikely event like region-wide outage, buckets in geo-redundant locations remain available, with no need to change storage path you cannot change bucket location or project after creation.

Lec.20. Google Cloud storage pricing and the concept of Bucket & Objects

- Cloud storage pricing is based on the following components:-
 1. Data storage
 2. Data Processing
 3. Network usage
- Data Storage-** The amount of data stored in your Buckets Storage rates vary depending on the Storage Class of your data and locations of your buckets.
- Data Processing-** The processing done by Cloud Storage, which includes operations Charges any applicable retrieval fees and inter-region replication
- Network Usage-** The amount of data read from or moved between your buckets.
For eg- Data storage costs in Iowa

	Standard	Nearline	Coldline	Archive
Regional	\$0.020	\$0.010	\$0.004	\$0.0012
Dual-Region	\$0.022	\$0.011	\$0.0044	\$0.0014
Dual-Region	\$0.026	\$0.010	\$0.007	\$0.004

- Note-** Dual regions are billed to both Underlying regions at the above prices For eg- Standard Storage in a dual region Comprised of Iowa and" Oregon will be billed at \$0.22 per GB per month for IOWA and \$0.22 per GB per month for Oregon.
- Data Processing Fees-** Data processing costs consist of operation charges, which apply to all requests made to cloud storage, retrieval fees, which apply to reading data stored in certain storage classes and Inter-region replication charges, which apply to data written to dual regions and multi-regions.
Price is \$0.05/10,000 operations in standard Storage, & \$0.10/10,000 operations in nearline and coldline
- Retrieval / Network Usage fees -**
This applies when you read, copy, move or rewrite object data or metadata that is stored using nearline storage, coldline storage or archive storage

Standard	Nearline	Coldline	Archive
\$0/GB	\$0.01/GB	\$0.02/GB	\$0.05/GB

- Default inter-region replication is free turbo replication is \$0.04/4B Chargeable.
 - > Data move within the same location is free
 - > Data moves different locations and the same continent is chargeable
 - > Data moves between different continents in chargeable
- **Bucket**
 - > Bucket is Like a folder for your objects
 - > Using buckets - You can organise and maintain access permissions
 - > You cannot create nested Buckets
 - > Bucket has a name, default storage class, and Geographic locations.
 - > Name and location can only be changed by deleting and recreating.
 - > There is only one Google Cloud storage namespace, which means every bucket must have a unique name across the entire Google Cloud storage namespace.
 - > Object names must be unique only within a given bucket.
- **Object**
 - > Object is file that you want to store.
 - > Can be upto 5 TiB.
 - > Object has data & Metadata.
 - > No limit on number of objects in Buckets.
 - > Objects are immutable and you can overwrite objects.
- **Turbo Replication**
 - > Turbo replication is a cloud storage premium feature designed to asynchronously replicate newly written Cloud Storage objects associated with any insert, rework, copy or compose operation- regardless of object size to a separate region within a target of 15 minutes.

Lec.21. Complete demo on Cloud Storage using GUI and Cloudshell

1. Login to GCP console ---> go to cloud storage --> Bucket --> Create Bucket
2. Name your bucket --> labels (optional)
3. Choose where to store your data
 1. Multi region
 2. Dual-region
 3. Region
4. Choose a default storage class for your data
Standard, nearline, coldline, archive
5. Choose how to control access to objects
Uniform, fine-grained
6. Choose how to protect object data protection tools
None, object versioning, retention policy
7. Data encryption
 1. Google managed encryption key
 2. customer-managed encryption key

Cloud storage demo using cloud shell

Open Cloudshell --> select your project

ls -a \$HOME

sudo rm -rf \$HOME ---> restart cloud shell

Create a bucket

gcloud mb -b on -l us-east1 gs://tgbucket454/

upload an object into your bucket

cloudshell--> : upload --> file --> upload file

gcloud cp local.gile gs://bucket

Download the object from your bucket

gcloud cp gs://bucket/file.jpg /home/download.jpg

copy the object to a folder in the buckert

gsutil cp gs://buckert1/file gs://tgbucket2/

list contents of a bucket or folder

gsutil ls gs://tgbucket2/

list details for an object
gsutil ls -l gs://bucket/file

make the object publicly accessible
gsutil iam ch allUsers:objectViewer gs://tgbuckt

Delete an object or bucket
gsutil rm gs:bucket

Delete entire bucket and its objects
gsutil rm -r gs://buckets

22. Object Versioning and Lifecycle Management

To support the retrieval of objects that are deleted or replaced, cloud storage offers the object versioning features.

Once enabled Object Versioning -

- Cloud Storage retains a **noncurrent object version** Each time you replace or delete a **Live object version**, as long as you do not Specify the generation number of the Live Version.
- Non-current versions retain the name of the object but are uniquely identified by their generation number.
- Non-current versions only appear in requests that explicitly call for them to be included.

If you disable object versioning -

- The bucket no longer accumulates new non-current Versions of objects.
- Object Versions that already exist in the bucket are Unaffected.

gsutil versioning set off gs://bucketname
gsutil versioning set on gs://bucketname
gsutil versioning get gs://bucketname
gsutil ls -a gs://bucketname
gsutil cp gs://bucket/file1 gs://bucket/file2
gsutil rm gs://bucket/file1

Lifecycle Management

Object Lifecycle Management

- To support common use cases like setting a TTL for objects, retaining non-current versions of objects, or downgrading storage classes of objects to help manage costs, cloud storage offers the object lifecycle management features.
- In order to use object lifecycle management, you define a lifecycle configuration, which must be set on a bucket. The configuration contains a set of **rules** which apply to current and future objects in the bucket. When an object meets the criteria of one of the rules, cloud storage automatically performs a specified **action** on the object.

Here are some examples of use cases

- Downgrade the storage class of objects older than 365 days to coldline storage.
- Delete objects Created before 1 Jan 2022.
- Keep Only the 3 most. recent versions of each object in a bucket with Versioning enabled.

Each lifecycle management configuration contains a set of rules. Each rule contains One action and one or more conditions.

- An object has to match all of the conditions specified in a rule for the action in the rule to be taken
- If you specify multiple rules that contain the same action, the action is taken on an object when that object matches the conditions in any of the rules.
- If multiple rules have their conditions simultaneously for a single object, cloud storage performs the action associated with on one of the rules, based on the following can migration.
- The delete action takes precedence over the SetStorageClass action.
- The SetStorageClass action that switches the object to the storage class with the lowest at-rest storage pricing takes precedence.

Create rule.json file and upload on gcp machine

```
{
  "lifecycle": {
    "rule": [
      {
        "action": {
          "type": "SetStorageClass",
          "storageClass": "NEARLINE"
        },
        "condition": {
          "age": 365,
          "matchesStorageClass": [
            "STANDARD"
          ],
          "matchesSuffix": [
            ".jpg",
            ".png"
          ]
        }
      },
      {
        "action": {
          "type": "SetStorageClass",
          "storageClass": "COLDLINE"
        },
        "condition": {
          "age": 1095,
          "matchesStorageClass": [
            "NEARLINE"
          ],
          "matchesSuffix": [
            ".jpg",
            ".png"
          ]
        }
      }
    ]
  }
}
```

gsutil lifecycle set rule.json gs://bucketname

Lec-23. How to Create Signed URL

- A Signed URL is a URL that provides limited permission and time to make a request.
- Signed URLs Contain authentication information in their query string, allowing users without credentials to perform specific actions on a resource.

Key parameters in sign URL

1. **X-Goog-Algorithm** - The algorithm used to sign the URL.
2. **X-Goog-Credentials** > Information about the credentials used to create the Signed URL
- 3 **X-Goog- Date** - The date and time the signed URL became usable, YYYYMMDD 'T' HHMMSS'Z'
4. **X-G1o0q-Expires** - The length of time the signed URL remained valid, measured in sec In our video, we will take 60 Sec. Max duration is 60,4800 Sec ie 7 days.
- 5 **X- Goog-SignedHeader** - Header that had to be included as part of any request that used the signed URL X-Goog-Signature - The authentication string that allowed request using this signed URL to access resources.

gsutil signurl -d 60s key.json gs://bucketname/imagefile.JPG

https://storage.googleapis.com/bucket5654/image.JPG?x-goog-signature=3999d1ef7267a689e5c2fd313cd419fe5ec37cea23664b1ae162f25b78004522b131e69e3b649c87b54d46faf0b54ce4c7e8ee086c8b793697ca0b4a3fcb95e9cfc1d93d3569a534944e4a7d76f6b61f38442ff5bc3a22ef6c03492f1900367da8255a8255153db2a4700803a76df0fa268073a667163974ee173636e8885e4c404ddb5a82a3c2d849b3dfdd643f95a52b444e73377c0171e56425d04e11c51351dbcb55517a2b6acba332f2019bd075c8e5d7a39d48a0f803c850d115701b5e506304a628c1bf39622ea212392712e9e76239ed2aa180566f11b22b6ae3122c47398dfe9c94fca420c93110b1c73d9c2c8d3460b50faba0192b61e69bf26e8&x-goog-algorithm=GOOG4-RSA-SHA256&x-goog-credential=682258878608-compute%40developer.gserviceaccount.com%2F20220901%2Fus%2Fstorage%2Fgoog4_request&x-goog-date=20220901T091459Z&x-goog-expires=60&x-goog-signedheaders=host

Lec 25. Google Cloud VPC Network

- VPC Network is just like your physical network except that it is virtualised within Google Cloud.
- A VPC Network is a global resource that consists of a list of regional virtual subnet works in data centers, all connected by a global wide area network.

A VPC Network provides -

1. Provides Connectivity for your VM, including GKE & app engine instances.
2. Offers Native Internal TCP/UDP Load Balancing.
3. Connects to **on-premises** network using **Cloud VPN Tunnels** and Cloud Interconnect attachments.
4. Distributes traffic from Google Cloud external load balancers to backends.
5. VPC always lies inside a project. One project can contain multiple VPC n/w.
6. New project start with a **default VPC** that has **one subnet in each region**.

VPC Networks have the following properties -

- VPC networks, including their associated **routes** and **firewall rules**, are **global resources** They are not associated with any particular region or zone.
- **Subnets are regional resources**. Each subnet is associated with a region. You can create **more than one subnet per region**.
- Each subnet has having unique network range inside a project.
- Inside and outside traffic can be controlled by firewall rules.
- Resources within a VPC, can communicate with one another by using IPv4 and IPv6 addresses.
- An Organisation Can use shared VPC to keep a VPC network in a Common host project
- VPC networks Can be Connected to Other VPC networks in different Projects or Organisations by using VPC network **Peering**.
- VPC Networks do not support Broadcast or Multicast addresses within the Network
- **Auto Mode** VPC networks Create Subnets in each region automatically.
- You **cannot connect** directly Auto mode VPC network to one another because they use the same ranges 10198.0.0/9

Lec.28- VPC Peering

- Google Cloud Network peering allows Internal IP address Connectivity across two VPC networks regardless of whether they belong to the same project/organisation or not.
- In VPC Network peering, traffic stays **within Google's Network** and doesn't traverse the public Internet.

KEY FEATURES

- VPC network peering works with **Compute Engine, GKE and App Engine** flexible environment.
- Each side of a peering association is set up independently peering will be active only when the configuration from **both sides** matches.
- Either side can choose to delete peering association at any time.
 1. VPC Network peering supports IPv4 connectivity only
 2. CIDR range cannot overlap in two peered Networks.
 3. Transitive peering is not supported.
 4. One VPC can directly peered with max 25 VPC.
 5. Peering is allowed with shared VPC.

Lec 29. Cloud NAT Gateway

- **Network Address Translation (NAT)** is a method of remapping one IP address space into another by modifying network address information in the IP Header of Packet. Usually, this is done in a NAT Device.
- IP Masquerading or NAT is a technique that hides an entire IP address, Usually the Private IP addresses, behind a single public IP. The hidden addresses are changed into a Single public IP as the source of the outgoing IP packet so they appear as originating from the routing devices and not from the hidden cost.
- Cloud NAT allows outbound traffic but restricts inbound traffic from external sources.
- Cloud NAT allows your resources to create outbound connections to the Internet without giving public IP to them.
- You can have **Multiple Cloud NAT** Gateways in the **same region** of a VPC but, Each gateway is configured for a different Subnet.

Cloud NAT Gateway provides outgoing connectivity for the following resources -

1. Compute Engine **VM instance** without external IP addresses.
 2. Private **GKE** Clusters.
 3. **Cloud Run** instances through serverless VPC access.
 4. **Cloud Function** instances through serverless VPC access.
 5. App Engine standard environment instances through serverless VPC access.
- Cloud NAT cannot apply to any custom route whose next hop is not the default Internet Gateway.
 - Google Cloud NAT is based on Andromeda ie Google Network virtualisation stack.
 - There is no intermediate NAT proxies in the data path Each VM is programmed by GCP to NAT using assigned ports Each Cloud NAT gateway is associated with a Single VPC network, and region and Cloud Router cannot use the same Cloud NAT in two or more regions.
 - You cannot use Cloud NAT for communication between VMs in a VPC Network. it is designed to provide connectivity to the Internet.

Cloud NAT Pricing

Cloud NAT Pricing is based on the following usage

- An hourly price of NAT Gateway that is based on the number of VM instances that are using the gateway.
- Per- GB Cost for ingress and egress data that is processed by the gateway.

Lec 30- Theory of Load Balancer in GCP - part 1

- A Load balancer distributes user traffic across multiple instances of your application By spreading the Load, Load Balancing reduces the risk that your applications experience performance issues.
- Cloud Load Balancer is built on the same frontend- serving Infrastructure that powers Google It supports 1 million+ Queries per second with consistent high performance and low latency traffic enters Cloud Load Balancing through 80+ distinct Global load Balancing locations, maximizing the distance travelled on Google's Past private n/w backbone.

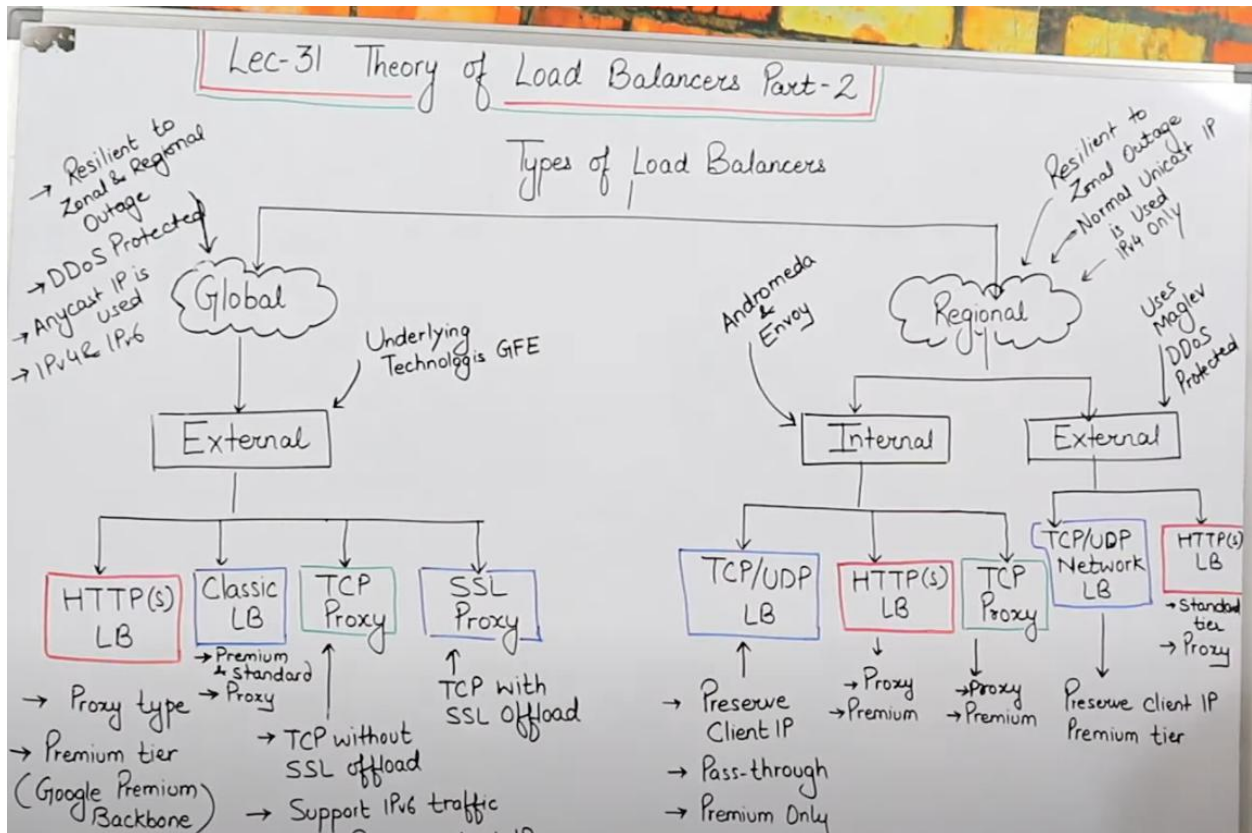
Google Cloud offers the following Load Balancing features -

1. Single anycast IP address
 2. Software-defined Load Balancing
 3. Seamless Autoscaling No Prewarming
 4. Layer-4 and Layer-7 LB
 5. External and Internal LB
 6. Global and Regional LB
 7. Enable Cloud CDN for HTTP(S) Load Balancing to Optimize application delivery for your users with a Single Checkbox.
 8. Manage SSL Certification and Decryption
- **Load Balancer can be opted on the Basis of your Requirement In Google, you have various aspects**
 1. External vs Internal LB
 2. Global vs Reginal
 3. Premium vs Standard Tier
 4. Proxy versus Pass- through
 5. Traffic Type
 6. DDoS Protection

Lec 30- Theory of Load Balancer - Part 2

- **Global load Balancing** is when your backends are distributed across multiple regions
- **Regional Load Balancing** is when your backends are in one region or when you have jurisdictional compliance requirements for traffic to stay in a particular region.
- **External Load Balancers** distribute traffic coming from the internet to your Google Cloud VPC.
- **Premium tier** delivers traffic on Google's premium Backbone. It gives high performance and low latency.

- **Standard tier** uses a regular ISP network. It is low-cost alternative for apps that don't have strict req. for latency & performance.
- **Proxy Load balancers** terminate incoming client connections and open new connections from the LB to the Backends.
- **Pass-through LB** do not terminate client connections, Instead, load-balanced packets are received by backend VMs with the packet source, destination, and if applicable, port information Unchanged connections are terminated by the backend VM's. Responses from the Backend VM's go directly to the clients, not back through the Load Balancer. The term for this is direct Server return.



Underlying Technology of Google Cloud Load Balancers

- Google Front Ends (GFEs) are Software-defined, distributed systems that are located in Google points of presence (PoPs) and perform Global Load Balancing in conjunction with Other Systems and Control Plane.
- Andromeda is Google Cloud's SDN virtualization Stack.
- Maglev is a distributed System for Network Load Balancing.
- Envoy Proxy is an Open Source edge and Service Proxy, designed for Cloud- Native applications.

Lec 34. Identity and Access Management Part-1

- IAM lets you grant granular access to specific Google Cloud resources and helps prevent access to other resources.
- IAM lets you adopt the Security principle of Least Privilege, which States that nobody should have more permissions than they actually need.
- With IAM, you manage access Control by defining who (Identity) has What access (role) for which resources.
- IAM, permission to access a resource is not granted directly to the end User Instead, permissions are grouped into roles, and roles are granted to authenticated Principals (Members).
- Policy defines and enforces what roles are granted to Which principals Each allow policy is attached to a resource.
- When an authenticated principal attempts to access a resource, IAM Checks the resource's allow policy to determine Whether the action is permitted.

Feature of IAM

- Lets you authorize who Can take specific actions on resources to give you full Control and Visibility on your Google Cloud Services Centrally
- Permissions are represented in the form of *service.resource.verb*
- Can map job functions into groups and roles.
- With IAM, users only get access to What they need to get the job done
- Cloud IAM enables you to grant access to Cloud resources at fine-grained levels, well beyond project-level access.

Basically, IAM has three important parts:-

Principal /Members - A principal can be a

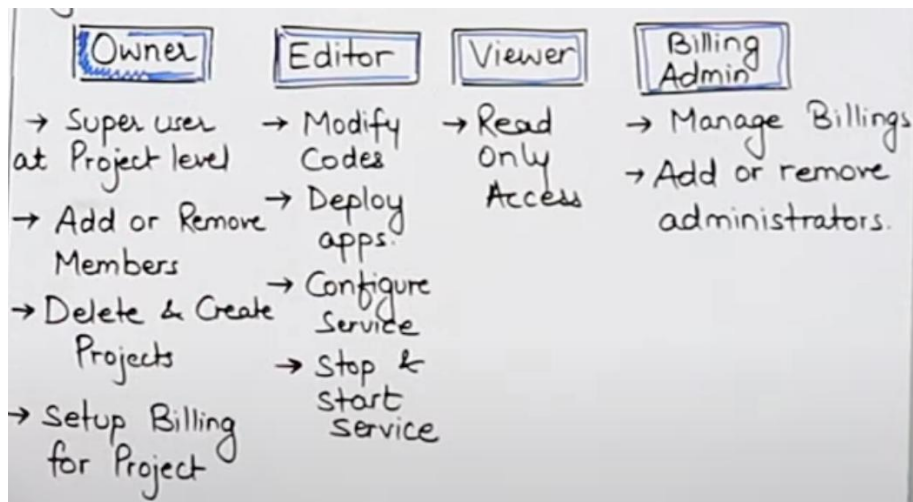
1. Google account
 2. Service account
 3. Google Group
 4. Google Workspace
 5. Cloud Identity Domain
 6. Authenticated Users
 7. All Users
 - 8.
- **Role** - A role is a collection of permissions. Permission determines what operations are allowed on a resource when you grant a role to a principal, you grant all the permissions that the role contains.
 - **Policy**- Policy is a collection of role bindings that bind one or more principals to individual roles when you want to define who has what type of access on a resource, you create allow policy and attach to the resource.

Lec 35. Identity and Access Management Part-2

- In IAM, you grant access to principals. Principals can be of the following types
 1. Google account
 2. Service account
 3. Google Group
 4. Google Workspace
 5. Cloud Identity Domain
 6. Authenticated Users
 7. All Users
- 1. **Google account** - Any email address that's associated with a Google account Can be an identity, including gmail.com or other domains
- 2. **Service account** - A Service account is an account for an application or compute Workload instead of an individual end User.
- 3. **Google Groups-** is a Collection of Google accounts and service accounts. Every Google Group has a unique email address that's associated with the group. You Can grant and change access controls for a whole group at once instead of granting or Changing access controls one at a time for individual users or service accounts.
- 4. **Google Workspace account** - Google Workspace accounts are associated with your organisation. Internet domain name, such as example.com. When you create a Google account for a new user, such as username@example.com, that Google account is added to the virtual group for your Google Workspace a/c.
- 5. **Cloud Identity Domain-** A Cloud Identity domain is like a Google Workspace account, because it represents a virtual group of all Google accounts in an organisation However cloud Identity domain users don't have access to Google Workspace applications and features.
- 6. **All Authenticated User** - The Valve of all Authenticated Users is a special identifier the represents all Service accounts and all users on the internet who have authenticated with a Google account. This identifier includes accounts that are not connected to a Google workspace. account or Cloud Identity domain, Such as personal Gmail accounts. Users who are not authenticated, such as anonymous visitors, are not included.
- 7. **All Users** - The Valve of all users is a special identifier that represents anyone who is on the internet / including authenticated and unauthenticated users.
- A role Contains a Set of permissions that allows you to perform specific actions on Google Cloud Resources.
- You don't directly grant users permissions in IAM. Instead, you grant them roles, Which bundle one or more permissions.
- To make permissions available to members, including users, group and Service accounts, you grant roles to the members.

There are three types of roles in Google Cloud IAM:-

- Basic /Primitive roles: -IAM Basic roles offer fixed, Coarse-grained levels of access



Predefined Roles

- Provides granular access for a specific Service and is managed and defined by Google Cloud.
- Prevents Unwanted access to Other Resources.
- Google is responsible for updating and adding permissions as necessary.
- You Can grant multiple roles to the same user.

Custom Roles

- Provide granular access according to a User-defined list of permissions
- You can create a Custom IAM role with One or more permissions and
- then grant that custom role to Users or Groups
- Custom roles are not maintained by Google
- You can grant multiple roles to a User or a group.

Policy

- A policy is a Collection of bindings, audit Configuration and metadata
- A binding associates one or more members with a Single role and any Context - specific Conditions that Change how and when the role is granted
- Each binding includes the following fields
- A member, Known as an Identity or Principal, Can be a:
 - User account
 - Service account
 - Google Group
 - Domain

- A role which is named Collection of permissions that grant access to perform actions on Google Cloud Resources.
- A Condition, which is a logical expression that further constraints the role binding based on attributes about the Request, Such as its Origin, the target Resource etc.
- You Can Grant roles to Users by Creating an Allow policy, which is a Collection of statements that define who has what type d access.
- A Service account is a special kind of account used by an application or a Virtual Machine (VM), not a person
- Applications use Service accounts to make authorized API Calls, authorized as either:
The service account itself
 1. as Google Workspace
 2. as Cloud Identity users through domain-wide delegation.
- A Service account is identified by its email address, which is unique to the account saname@projectid.iam.gserviceacc.com
- Each Service account is associated with two sets of public/private RSA Key pairs used to authenticate to Google.

Types of Service Accounts

1. Default Service account
 2. User-managed Service accounts
- A Service account is also a resource with IAM policies attached to it, which means you can define who can use the account and who can perform specific actions on the service account.
 -

Lec.37- Google Cloud IAM Demo Part -2

- A Service account is a Special kind of account Used by an application or Compute Workload, Such as a compute engine VM instance, rather than a person.
- for eg - A Service account Can be attached to a Compute Engine VM So that applications running on that VM Can authenticate as the Service account. In addition, the Service account Can be granted IAM roles that let it access resources.

Difference between Service account & User account -

- Service a/c do not have passwords, and cannot log in via browsers or cookies.
- Service accounts are associated with Public/Private RSA Key pairs that are used for authentication to Google, and for signing data.
- You can let other users or service a/c impersonate a Service a/c.
- Service accounts do not belong to your Google Workspace domain, unlike user accounts.

Type of service account: 1. Default 2. User Managed 3. Google managed

1. Default service account:

- When you enable or use some Google Cloud Service, they create default service accounts that enable the service to deploy jobs that access other Google Cloud resources. These accounts are Known as Default service accounts
- these two Services Greats Service accounts are App Engine, Compute Engine and any Google Cloud Service that Uses App Engine or Compute Engine.

2. User-managed service account -

- You can create a user-managed service account in your project using IAM API, Console and CLI.
- By default, you can create 100 user-managed service accounts in a project.
- When you create a User-managed service account in your project, you choose a name for the Service account. Which user the following format:
- `service.account.name@project-id.iam.gserviceaccount.com`

3. Google managed Service account

- Some Google Cloud Services need access to your resources so that they can act on your behalf for eg when you use Cloud Run to Run a Container, the Service needs access to any pub/Sub topics that Can trigger the Container.
- To meet this need google Created a Google managed service account not listed in the service account page.

Types of Keys for Service Accounts

1. Google Managed Key pairs are used by the Service account Credentials API, and by Google Cloud Services Such as App Engine & Compute engine.

2. User- Managed Key pairs

You Can Create User-managed key pairs for a service account, then use the private key from each key pair to authenticate with Google APIs This private key is known as a service account Key Each Service account can have up to 10 S.A. Keys.

The user is responsible for the Security of the private key.

Lec.38- Google Cloud IAM Demo Part -3

Type of roles in GCP

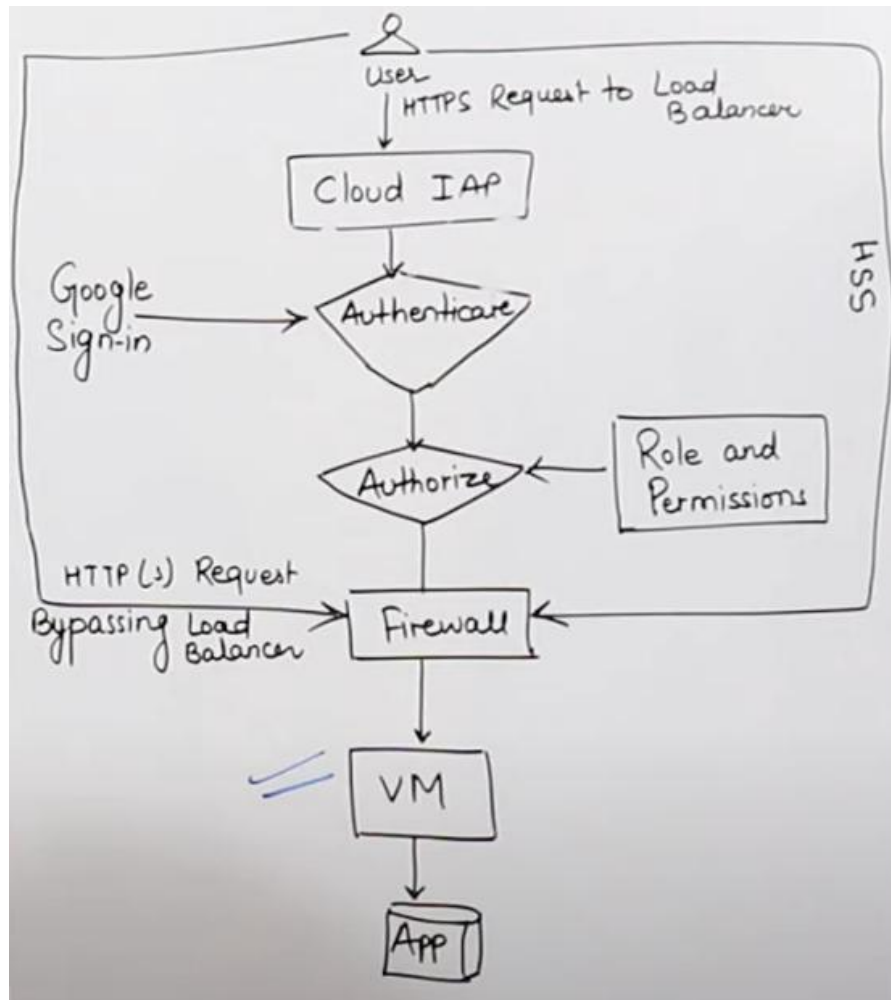
1. Basic
2. Predefined
3. Custom role

Lec 39- Identity Aware Proxy- Demo

- IAP lets you establish a Central authorization Layer for applications accessed by HTTPS, so you Can use application-level access Control model instead of relying on network-level firewalls.

How IAP Works

- When an application or resource is protected by IAP, it Can only be accessed through the proxy by principals also Known as users, who have the Correct identity and access Management role.
- When a user tries to access an IAP-secured resource, IAP performs an authentication and authorization check.



- IAP does not support or use VPN.
- IAP allows access to web apps from anywhere.
- IAP allows the Zero-trust Security model.
- IAP supports Google Workspace accounts.
- IAP works for app engine, Cloud run, compute engine, GKE and on-premises.
-

Lec-40. App Engine- Standard & Flexible

- The App Engine is Google's PAAS offering It is a Compute Service that provides a managed platform for running applications.
- As this is a managed service, your focus Should be on the application only and Google will manage the resources needed to run the application.
- It was GCP earliest Service, introduced in 2008.
- Applications in the app engine run across multiple Servers
- No Usage Charges - pay for resources provisioned

Features of App Engine

- App Engine makes deployment, maintenance, and Scalability easy - so that the user may focus on innovation.
- Automatic Load Balancing and Scaling.
- Application Versioning and Traffic Splitting.
- Supports Several Programming Languages like **Go, Python, Java, NodeJS, Net & Ruby**.
- Suitable for Web applications.

Features	Standard	Flexible
Instance startup	Seconds	Minutes
SSH Access	No	Yes
Write to local disk	No (Some runtime have read and write access to the /tmp/directory)	Yes, Ephemeral (disk initialized on each VM Startup), Can access app engine Resources
Need of VPC	No	Yes
Pricing Model	After free daily use, pay per instance Class, with automatic Shutdown	Pay for Resource allocation per hour
Scaling to zero	Yes	Min 1 instance
Scaling	Manual, Basic & Automatic(F)	Manual & Automatic
	Sandbox environment	App run in Docker container, support more languages

	Compute Engine	App Engine
Service model	IaaS	PaaS
Type of service	Unmanaged	Managed
Control of resources	More control and flexibility over resources	Less control over computing resources
Cost	less	high
Running Instances	Min 1	scale to zero
Autoscaling	Slower autoscaling	Faster autoscaling

Flexible

- The Flexible environment provides more options and Control to the developers who want to use App Engine, but without the Language Constraints of the standard Environment. It uses Docker Containers as the basic building blocks. These Containers Can be auto-scaled according to load.

Lec-43. Cloud Run

Cloud Run is a managed compute platform that lets you run containers directly on top of Google's Scalable infrastructure

- Launched in 2019
- AWS Fargate and Azure Container instances are Equivalent Services to Cloud Run.
- You can deploy code written in any programming language.
- Containers can be called through Web requests or Pub/Sub Events.
- Cloud Run offers automatic scaling.
- Cloud Run provides for an open API and runtime environment.
- Cloud Run comes with a free-tier with additional Billing done on a pay per use basis.
- It is serverless and runs stateless containers.

Two ways to run your Code-

On Cloud run, your Code Can either run Continuously as a Service or as a job.

1. **Cloud Run Services**- Used to run Code that responds to web requests or events.
2. **Cloud Run Jobs** - Used to run code that performs work (a job) and quits when the work is done A Cloud Run Service provides you with the infrastructure required to run a reliable HTTPS endpoint. Your responsibility is to make sure your code listens on a TCP port and handles HTTP Requests.

Features

1. **Unique HTTPS Endpoint_for every Service:** Every Cloud Run Service is provided with an HTTPS Endpoint on a Unique Subdomain of the *.run. appdomain and you Can Configure custom domains as well.
 2. **Fast request-based autoscaling**
 3. **Built-in Traffic Management:** Every deployment Creates a new immutable revision You Can route incoming traffic to the latest Version, roll- back to the Previous version, or split traffic to multiple revisions at the same time.
 4. **Private & Public Services**
- There are two pricing Models
 1. Request based - if a container instance is not processing requests, the CPU is not allocated and you are not charged.
 2. Instance-based - You are charged for the entire lifetime of a Container instance and the CPU is always allocated There is no per-request fee.

Lec 45- Google Cloud Function

- With Cloud Functions you write simple, single-purpose functions that are attached to Events emitted from your cloud infrastructure and Services.
- A pay-as-you-go function as a service (FaaS) to run your Code with Zero Server Management
- Cloud functions Can be written using :
 1. Node.Js
 2. Go
 3. Python 3
 4. Java
- Automatically Scales based on load without thinking about the infrastructure.
- Built-in Security at the role and per function level based on the least privileged principle.
- Allow you to trigger your code from Google Cloud, Firebase, and Google Assistant or Call it directly from any web, mobile or backend application via HTTP
- To act on events, you shall define a trigger. Binding a function to a trigger enables you to act on events. The most common triggers are-
 1. Google Cloud Storage
 2. Pub/Sub
 3. HTTP
 4. Firestore

Pricing - GCF pricing account to

- How long your function runs
- How many times it is invoked
- How many resources you provision for the function

Differences between GAE, GCF, GCR

	App Engine	Cloud Functions	Cloud Run	Cloud Run for Anthos
Deployment type	App	Functions	Cloud Run	container
Scaling	yes	yes	yes	Only using pods
Runtime support	Java, Node js, Python, Go, PHP	Node.Js, Python, Go, Java	All Available	All Available
Access control	Auth 2.0, CIGD, firebase Auth, Google sign-in	IAM Permission	IAM Permission CIGD, Google sign-in. Firebase Auth	Cluster-only VPC-only
HTTP/gRPC	No	No	No	Yes
Custom domains	Yes	No	Yes	Yes
GPU/TPU Support	No	No	No	Yes
Timeout	1 Min	9 Min	15 Min	A5 Min
VPC Networking	Yes	Yes	No	Yes

Lec 46- Google Cloud SQL

- Cloud SQL is a fully Managed Relational Database Service for MySQL, Postgre SQL and SQL Server.
- Cloud SQL is fully hosted on Linux.
- Cloud SQL is a Web Service that allows you to create, configure, configure and Use relational databases that live in Google's Cloud.
- Any Existing databases in MySQL, Postgre SQL or any other structured database can be migrated to Cloud SQL.
- Google handles replication, patch management and database management to ensure availability and performance.
- Fully Managed MySQL (5.6,57 &80), Postgre SQL (96, 10, 11, 12, 13 & 14) & SQL Server (2017, 2019)
- Easily scale up to 96 vCPO and 624 GB & RAM(Vertical Scalable)
- Upto 64 TB of Storage available
- Support for private IP
- Data replication between multiple zones with automatic failover
- Import and export databases using mysqldump, or import and export CSV files
- Instance Cloning

To use cloud SQL effectively, you should understand some of the Concepts on which it is built

1. Cloud SQL Instance -> A Cloud SQL instance Corresponds to one VM. The VM includes the database instance and accompanying software contained to keep the database instance up and running.

2. Database instance - A database instance is the Set of software and files that Operate the databases: My SQL , Postgre SQL or SQL Server

3. Public & Private IP - In cloud SQL, public IP means that the instance is accessible through the public internet, but are accessible through a VPC. Cloud SOL instances can have both a public and a private IP.

4. High Availability > HA in cloud SQL works by having two sync. instances: Primary and standby. Each instance has exactly one VM. Each instance is in a different zone in the Same Region.

5) Failover > A failover is when cloud SQL Switches from the Original primary instance to the standby instance.

6) Standby instances - The Standby instance is used in H.A to replace the primary instance when failover occurs. The Standby instance doesn't appear in the Google Cloud Console. When failover occurs, connections to the primary instances are automatically transferred to the Standby instance.

7) Clone- When you Clone a Cloud SQL instance, you Create a new instance that is a Copy of the Source instance, but is Completely independent. After Cloning is Complete, Changes to the source instance are not reflected in the Clone, and Changes in the Clone are not reflected in the source instance.

8) Replication- Replication is the ability to create Copies of a Cloud SQL instance or an on-premises database, and offload work to the copies. The main reason for using replication is to scale the use of data in a database without degrading performance on the primary instance.

Database Administration

- Cloud SQL lets you Create and delete databases and database users, but it is not a database administration tool. There are many database administration tools you can choose from depending on your database engine, including the following:
 1. PHPMyAdmin
 2. MySQL Workbench
 3. Toad Edge
 4. SQL Server/Management studio

Cloud SQL Pricing

- How much Storage you provision
- How many vCPU you select
- How much RAM you Select
- Where to Choose to host your data
- How much network traffic leaves your instance
- How many IP addresses you use

Cloud SQL Updates

- Configuration Updates
- System Updates

Lec-48 Cloud spanner

- A fully managed Relational Database Service with Unlimited Scale, Strong Consistency, mission-critical and Upto 99.999% Availability.
- In Aws, AWS Aurora is Available.
- Cloud Spanner launched in 2017.
- Provide transparent, Synchronous replication across region and multi-region Configurations
- Optimizes performance by automatically Sharing the data based on request load and Size of data so you can spend less time thinking about scaling your database and more to scaling your business.
- You can run instances on a regional Scope or multi-regional where your database is able to survive regional failure.
- Used for Gaming, Logistics, e-commerce, Insurance etc.

	Cloud Spanner	Traditional Relational	Traditional non Relational
Schema	Yes	Yes	No
SQL	Yes	Yes	No
Consistency	Strong	Strong	Eventual
Availability	High	Failover	High
Scalability	Horizontal	Vertical	Horizontal
Replication	Automatic	configurable	configurable

Pricing

- Number of nodes in your instance
- Amount Storage that your tables and Secondary indexes uses.
- Amount of network bandwidth used.
- No additional charge for Replication

Limitations d Cloud Spanner

- Lift and Shift
- Lots of in-database business logic (triggers and stored procedures)
- When you need less latency

Lec-50 Bigtable

- Cloud Bigtable is a fully managed, wide-column **No-SQL** Databases that offers low latency and replication for high availability.
- Single-digit millisecond latency, limitless Scale and 99.999% availability for large analytical and Operational Workloads.
- Easily migrate from Apache HBase & Cassandra to Bigtable with no downtime.
- Bigtable is a sparsely populated table that Can Scale to billions of rows & thousands of columns.
- Bigtable supports storage of terabytes or even petabytes of data.
- Bigtable is not a relational database It dose not support SQL queries, joins or multi-row transactions.
- Data replication can be performed by simply adding a second cluster to the instance and replication starts automatically.
- Bigtable integrates easily with big data tools like Hadoop, Dataflow, Dadaproc and supports HBase APIs.
- Bigtable Stores data is Colossus, Google's file system.

Instance

- A Bigtable instance is a container for data.
- Instances have one or more clusters, located in a different zone and different region.
- Each Cluster has at least 1 node.
- A Table belongs to an instance and not to the cluster or node.
- An instance also Consists of the following properties.
 1. Storage type: SSD or HDD
 2. Configuring cluster-ID, selecting region and zone
 3. Node scaling mode: manual or auto
 4. Encryption- GMEK, CMEK
 5. Add replication cluster

Cluster

- Cluster handles the requests sent to a Single Bigtable instance.
- Each Cluster belongs to a Single bigtable instance and an instance can have upto 4 Clusters.
- Each Cluster is located in a Single Zone.
- Bigtable instances with only 1 cluster do not use replication.

Nodes

- Each Cluster in an instance has 1 or more nodes, which are the compute resources that Bigtable uses to manage the data.
- Each node in the Cluster handles a subset of the request to the Cluster
- All Client requests go through a front-end Server before they are sent to a Bigtable node

- Bigtables separate the compute from the storage. Data is never stored in nodes themselves, each node has pointers to a set of tablets that are stored on Colossus.
- Recovery from the failure of a bigtable node is very fast as only the metadata needs to be migrated to the replacement node.
- when a bigtable node fails, no data is lost a bigtable cluster can be scaled out by adding nodes which would increase.
 1. the no. of simultaneous req. that the cluster can handle.
 2. the max throughput of the cluster.
- Bigtable nodes are sometimes referred as tablet servers.
- Bigtable has a limit of 1000 tables per instance.

Lec-52 Cloud Memorystore

- Memorystore is a fully-managed database Service that provides a managed Version of two popular Open source Caching Solutions Redis & Memcached.
- Memorystore for Redis provides a fully- managed Service that is powered by the Redis in-memory data store to build application Cache that provides Sub-millisecond data access.
- Memorystore for Redis offers Several advantages over self- managed Redis.
 1. Deploy what fits your needs.
 2. Easily scale to get blazing speed.
 3. Highly available & more Secure.
 4. Focus on your application
 5. Redis Protocol Compatible
- It is good for caching, gaming & stream processing

Features of memorystore for Redis

1. Fully managed
2. Simple deployment
3. High Availability
4. Enterprise Grade Security
5. Scale, as needed
6. Monitoring
7. On- demand Billing
8. Redis versions 6.X, 5.0,4.0 & 3.2

- You Can use any standard Redis Client on the following environments to Connect to your Memorystore for Redis Instance.
 1. App Engine Flexible/ environment standard
 2. Compute engine VM instance
 3. Cloud Function
 4. Cloud Run
 5. GKE Clusters

Spec.	Basic tier	Standard tier
Maz. Redis Node Size	300 GB	300 GB
Max. Network Bardwidth	16 Gbps	16 Gbps
Scale node Size	Yes	Yes
Cross-zone Replicaton	No	Yes
Automatic Paulover	No	Yes
Read- Replica	No	Yes

Read Replica Behaviour

- Read replicas are not enabled on standard tier instances by default.
- Once read replica is enabled, you can't disable it.
- Standard tier instances can have 1 to 5 read replicas.

Limitations

1. Read replicas are only supported for instance fixes with nodes ≥ 5 GB
 2. Read replicas can only be enabled on instances that use Redis version 5 or higher.
- Each instance has a primary endpoint and read endpoint. The primary endpoint always direct traffic to the primary node, while the read endpoint automatically load-balance read queries across available replicas.

Failover

- When a primary fails, the memystore health monitoring service initiates the failover
- and the new primary is made available for both read and writes. The failover usually complete in under 30 seconds.
- When a failover happens the primary endpoint automatically redirect traffic to the new primary. However, all client connections to the primary endpoint are disconnected during a failover application with connection retry logic will automatically reconnect once the new primary is online.

Zone Failure

- If the zone where the primary is located fails the primary automatically fails over to a replica in another zone. If the instance only has one replica, then the read endpoint is unavailable for the duration of the zone outage. If the instance has more than one replica, replicas outside of the affected zone are available for reads.
- If the zone where one or more of the replicas is located fails, those replicas are unavailable for the duration of the zone failure. If there is a two zone failure and there are two or more replicas, the replica with the least lag in the remaining zone is made primary.

Maintenance on Memystore for Redis

- Most of these updates are performed while your redis instance is up and running
- However, certain system updates require a brief service interruption to be performed. These updates are Called maintenance.
- Maintenance typically falls into the following Category:-
 1. Memystore features - To launch some features, memystore requires a maintenance Update.
 2. Operating system patches: We continuously monitor for newly identified security vulnerabilities in the O.S. Upon discovery, we patch the O.S to protect you from new risks.
 3. Database Version Upgrade:- Maintenance can include an OSS Redis patch minor version update which does not include breaking or incompatible Changes.
- The duration of the maintenance window is 1 Hour.
- If maintenance window is not Specified for your instance, maintenance can begin at any time without advanced notification.

Version Upgrade Behaviour

- Data is preserved during a Version Upgrade Operation for both Basic and Standard tier instances. During an upgrade Operation, the instance undergoes a short Connection reset of a couple of minutes or less. The IP address and Connection string remain the same.

Import & Export data

- Import & Export uses cloud storage buckets to store RDB files.
- An instance can only Import backups from older Redis Versions.

Import feature Behaviour

- For optimal performance, you should import from cloud storage buckets located in the same region as your redis instance.
- Your redis instance is unavailable during an import operation. Your instance becomes available after the import operation finishes.
- If the import is successful, the existing data in the instance is overwritten by the data in the RDB file.
- If the import fails, the instance is brought Online, but the data may be fully flushed. You Can retry the import using the same RDB file, or you may use another RDB file to restore data.

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